needs many lists of canopies to be transferred to the reducer, and a custom version of the canopy clustering algorithm that handles merging centroids.

Listing 13.4 summarizes a high-level implementation of canopy clustering leveraging MapReduce, with classification performed in the same method, but in a second MapReduce job with respect to the choice of centroids; lines #2-#5 run the same algorithm that in listing 13.3 performs the distributed computation of canopy's centroids; lines #6-#9, instead, run the code specific to this version, performing the assignment of each point p to all the canopies for which p is within a distance T_1+T_2 from the canopy's center: as we have mentioned, choosing a larger radius ensures that so that no points will remain uncovered.